

COLUMBUS AFB, MS, USA

WMO: 723306

Lat: **33.650N**Lon: **88.450W**Elev: **67**StdP: **100.53**Time Zone: **-6.00 (NAC)**Period: **94-19**WBAN: **13825****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.3	-4.1	-13.7	1.2	-2.3	-11.1	1.5	-0.9	10.7	9.8	9.5	7.4	3.0	340	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	35.8	24.5	34.6	24.8	33.5	24.7	27.4	32.2	26.7	31.7	25.9	30.9	3.3	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	21.8	29.8	25.3	20.6	28.7	24.8	19.9	28.3	86.6	32.1	83.4	31.7	80.4	31.1	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.7	6.9	-9.4	37.1	2.9	1.7	-11.5	38.3	-13.2	39.3	-14.9	40.2	-17.0	41.4		
(5)			-10.3	28.1	2.8	1.1	-12.3	28.8	-13.9	29.4	-15.5	30.0	-17.5	30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	6.5	9.6	13.2	17.4	22.1	26.0	27.3	27.1	24.2	18.0	11.4	8.6
	DBStd	8.51	6.04	5.89	5.29	4.04	3.38	2.27	1.93	2.05	3.24	4.49	4.76	5.44
	HDD10.0	384	144	73	30	2	0	0	0	0	0	2	39	94
	HDD18.3	1444	369	248	171	64	10	0	0	0	2	63	212	304
	CDD10.0	3181	35	62	129	225	376	478	538	531	426	249	82	51
	CDD18.3	1199	1	4	12	37	127	228	279	273	178	52	4	3
	CDH23.3	11970	1	26	117	372	1189	2297	2922	2779	1707	520	32	7
	CDH26.7	5316	0	1	17	79	438	1072	1413	1366	767	160	3	0
Wind	WSAvg	2.8	3.5	3.5	3.5	3.3	2.7	2.4	2.1	2.0	2.2	2.5	2.7	3.1
Precipitation	PrecAvg	1422	140	133	148	141	113	108	119	95	87	79	107	149
	PrecMax	1889	270	223	307	362	345	227	261	194	267	276	212	317
	PrecMin	814	67	42	18	60	28	10	28	21	2	0	45	43
	PrecStd	254	51	51	53	70	73	61	62	47	55	62	41	63
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.7	26.1	28.5	30.5	33.3	36.0	36.9	38.1	36.0	32.4	26.5	24.3
		MCWB	17.4	18.8	18.8	20.6	23.0	24.5	25.0	23.8	22.7	21.8	19.0	20.0
	2%	DB	20.6	23.8	26.4	28.9	32.1	34.6	35.2	35.9	34.1	30.2	23.8	21.8
		MCWB	16.8	17.9	17.7	19.8	22.6	24.5	25.7	24.3	23.5	21.5	17.7	18.6
	5%	DB	18.6	21.6	24.3	27.4	30.8	33.3	34.1	34.2	32.8	28.4	21.9	19.8
		MCWB	15.7	16.3	16.9	19.1	22.3	24.4	25.5	25.0	23.6	20.3	17.4	17.3
	10%	DB	16.5	19.1	22.3	25.6	29.5	32.2	33.0	32.9	31.3	26.5	20.0	17.6
		MCWB	13.4	14.9	15.9	18.1	21.6	24.1	25.3	25.0	23.1	19.3	15.9	15.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.5	21.0	21.2	23.0	25.1	27.2	28.5	28.6	26.2	25.0	21.9	20.9
		MCDB	21.2	23.8	25.0	27.4	30.1	32.1	33.4	33.0	31.1	28.3	23.4	23.2
	2%	WB	17.9	19.3	19.7	21.7	24.1	26.3	27.4	27.6	25.2	23.5	20.1	19.1
		MCDB	19.4	22.3	24.6	26.2	29.3	31.6	32.3	32.2	30.7	27.9	21.9	21.0
	5%	WB	16.4	17.8	18.4	20.7	23.4	25.7	26.7	26.6	24.6	22.3	18.2	17.7
		MCDB	18.1	20.5	22.3	24.9	28.7	30.9	31.9	31.7	29.9	26.2	20.5	18.9
	10%	WB	14.4	16.0	17.3	19.7	22.7	25.1	26.0	25.8	24.0	21.0	16.7	15.7
		MCDB	16.2	18.3	20.6	24.0	27.6	30.1	31.2	30.5	28.9	24.2	19.2	17.4
Mean Daily Temperature Range	5% DB	MDBR	11.0	11.8	12.7	13.1	12.2	11.3	10.9	11.2	12.4	13.6	12.8	10.7
		MCDBR	12.3	13.8	14.9	14.3	13.6	12.9	12.3	13.1	14.0	15.0	14.5	11.8
		MCWBR	8.9	8.2	7.8	6.7	5.4	4.6	4.4	4.3	4.9	6.7	8.9	8.4
	5% WB	MCDBR	9.8	11.1	12.0	11.3	11.5	11.0	11.0	10.7	11.3	11.0	11.3	10.2
		MCWBR	8.3	7.7	7.7	6.4	5.1	4.7	4.8	4.7	4.6	6.0	8.6	8.2
Clear-Sky Solar Irradiance	taub	0.322	0.328	0.364	0.397	0.433	0.479	0.515	0.515	0.436	0.369	0.341	0.319	
	taud	2.513	2.491	2.413	2.317	2.260	2.151	2.082	2.076	2.286	2.453	2.472	2.537	
	Ebn at Noon	893	924	911	892	858	816	784	776	827	866	860	876	
	Edh at Noon	84	95	111	128	137	153	163	161	125	98	87	78	
All-Sky Solar Radiation	RadAvg	2.60	3.10	4.26	5.42	6.01	6.36	6.11	5.60	4.94	4.02	2.94	2.22	
	RadStd	0.22	0.43	0.40	0.39	0.42	0.46	0.34	0.37	0.42	0.52	0.29	0.22	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	+95

Nomenclature: See separate page